Background reading · The tree of life

Life Sciences · ages 11–14 · to read after the lesson

In class you met four mystery organisms and tried to sort them into three boxes. The sorting did not work, and that was on purpose. This sheet explains why, and what we did about it.

1 · The four organisms

	What it was	What it is made of	Has a nucleus?	Alive?
Mystery 1	The cholera bacterium	One single cell, about two thousandths of a millimetre long	No. Its DNA floats loose in a ring	Yes
Mystery 2	The rabies virus	Not a cell at all. A shell with a strip of genetic material inside. About ten times shorter than Mystery 1	No	Scientists disagree — see section 7
Mystery 3	The malaria parasite	One single cell, but a complicated one	**Yes**	Yes
Mystery 4	A tapeworm	Millions of cells. It can grow several metres long	Yes, in every cell	Yes

2 · The trick in the sorting

Look at Mystery 3 and Mystery 4. Both are parasites. One is a single cell. The other is an animal made of millions of cells, with a head and hooks.

Are they relatives? Not even slightly. They are about as different from each other as you are from a mushroom.

So here is the trick that was played on you. Two of the three boxes said **what a thing is**:

- **Bacterium** is a kind of living thing.
- **Virus** is a kind of thing.

But the third box was different:

- **Parasite** is not what something is. It is **what it does for a living**.

A parasite is any organism that lives on or inside another organism and takes what it needs from it. That is a job description, not a family. A parasite can be a single cell, an animal, a fungus — even a bacterium.

That is why the sorting kept feeling wrong. You were being asked to put things into boxes that were not the same kind of box.

3 · So what *can* you compare?

If lifestyle does not work, try looking at them instead.

Mystery 1 has a long tail for swimming. The tapeworm has no tail. The tapeworm has hooks. The bacterium has no hooks. One has no nucleus, one has one nucleus, one has millions.

There is nothing on the outside that all of them have. You cannot measure a bacterium against a tapeworm, because they do not share any parts to measure.

For about a hundred years this was a real problem in biology. You could compare two animals by their bones. You could not compare an animal with a bacterium at all.

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4 · The one machine every cell has

There is one thing.

Every living cell has to build proteins, and every living cell builds them with the same machine. It is called a **ribosome**.

Your cells have ribosomes. The cholera bacterium has them. The tapeworm has them. Every cell that anyone has ever looked at, anywhere, has them.

And a ribosome is partly built out of RNA — a long molecule written in letters, which means it can be **read**.

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5 · Where that machine came from

Stop and ask the question we asked in class. Every cell on Earth has the same machine. **Why would that be?**

If every kind of life had invented that machine separately, on its own, would they all have come out identical? No. Build something twice from scratch and you get two different things.

Identical means **inherited**, not invented.

So a very long time ago there was one population of cells that already had this machine. Everything alive today is descended from that population. It has a name: **LUCA**, which stands for the **Last Universal Common Ancestor**.

One thing LUCA is **not**: it is not the first living thing. It is the last ancestor that everything alive *today* still shares. There may well have been life before LUCA that left no descendants at all.

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6 · Reading the machine

In **1977** a scientist called **Carl Woese** had the idea that changed all of this. Stop looking at bodies. Read the machine.

He read one particular piece of ribosome RNA — it has a name, **16S** — from organism after organism, and compared the letters.

This works on anything that has a cell. The bacterium and the tapeworm, finally measured with the same ruler.

And now the counting makes sense. You and the cholera bacterium have this same gene because you both inherited it from LUCA. It has been slowly changing in your family and in its family ever since your two lines parted. So the differences are not random noise:

> **The number of differences is a measure of how long ago two organisms parted.**

7 · Why viruses are not on the tree

Go back to Mystery 2.

A virus has no ribosome. It does not have one because it is not a cell. There is nothing to read.

So Woese's ruler cannot measure it at all. That is why viruses are drawn **beside** the tree of life rather than on it.

This is also why the argument about whether a virus is alive did not get a clean answer. It is not that scientists know and nobody told you. **Scientists genuinely disagree**, and they disagree for exactly this reason: a virus has no machinery of its own. Some say that means it is not alive. Others say it is alive, but only by borrowing.

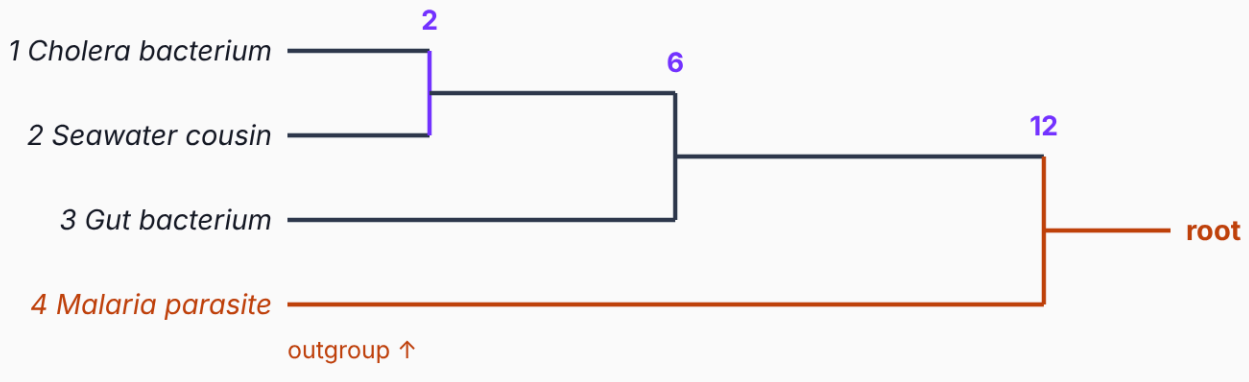
8 · What the numbers told you

In class you compared four short pieces of that same RNA.

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1  Cholera bacterium    GCTAACGAGCAGTGCAATCG
2  Seawater cousin      GCTAACGAGCAGTGTGATCG
3  Gut bacterium        GCTAACGAGTCAGCTAATCG
4  Malaria parasite     TACGTGATTCAGGCTAATCG
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	1	2	3	4
1	—	**2**	6	12
2	**2**	—	6	12
3	6	6	—	12
4	12	12	12	—

Drawn as branches, that comes out like this. Each fork is labelled with the distance at which it formed, and the further right a fork sits, the longer ago that split happened.



Read it from the left. Sequences 1 and 2 join first, at a distance of 2 — their fork is close to the tips, because they separated most recently. Sequence 3 joins them at 6. Sequence 4 joins last, at 12, and the root comes out of that final fork.

Notice where the root is. It is **not** on sequence 4. It sits **between** sequence 4 and the other three — that is the point where their two lines parted. Sequence 4 is not anybody's ancestor. It is a living organism, exactly as modern as the other three, and it has been changing for just as long. It only looks "deep" on the drawing because its branch joins the others further back.

In bracket form the same tree is written **((1, 2), 3), 4**.

Sequences **1 and 2** differ in only 2 letters out of 20. They are the closest pair, and they really are close relatives: both live in warm seawater.

Sequence **4** is **12 away from all three** of the others — the same distance to each, and the biggest number in the table. That makes it the **outgroup**: it branched off before the other three separated from each other.

And here is the part worth remembering. Three of those four are bacteria. Number 4 is not — it is the malaria parasite, and it has a nucleus. **The letters told you that before anyone mentioned nuclei.**

One last thing you may have spotted: the last four letters are the same in all four organisms. Some positions almost never change, because a change there breaks the machine and the cell dies. Those unchanging stretches are exactly what lets us line up organisms as different as a bacterium and a tapeworm.

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9 · Words to know

Ribosome — the machine inside every living cell that builds proteins.

16S rRNA — the particular piece of ribosome RNA that Woese read and compared.

LUCA — the Last Universal Common Ancestor: the population of cells that everything alive today is descended from.

Outgroup — the organism you already know is the most distant one. It tells you which end of a tree is the bottom.

Parasite — an organism that lives on or inside another one and takes what it needs from it. A way of living, not a family.

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10 · Three questions to think about

1. A new living thing is discovered at the bottom of the ocean. It has a cell, but it looks like nothing anyone has seen before. Which single molecule would you read first, and why that one?
2. The tapeworm has no mouth and no gut. It is still an animal, and a much closer relative of yours than the cholera bacterium is. Explain how both of those things can be true at once.
3. Suppose you compared sequences 1, 2 and 3 only, and left out the malaria parasite. You would still know that 1 and 2 are the closest pair. What exactly would you no longer be able to say?